

Project 22: Movie Revenue Prediction

Introduction to Artificial Intelligence

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Midterm Report

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1 Description of the Task

The primary objective of this project is to develop a machine learning model capable of predicting the global revenue of movies. Given a specific set of movie attributes such as budget, genres, original language, and release date. The model aims to output a precise financial prediction. We frame this project as a supervised regression task. Unlike a classification problem that would categorize movies into success levels, our goal is to predict a continuous numerical value that represent the total worldwide revenue in dollars. The challenge lies in the high variance of movie earnings and the need to capture non-linear relationships between production features and commercial success.

2 Description of the Dataset

The data for this project are sourced from Kaggle (*The Movies Dataset*). It provides a comprehensive set of metadata for several thousand films. From the available features, the following were selected as inputs to the model: `budget`, `genres`, `keywords`, `original_language`, `production_companies`, `production_countries`, `release_date`, `runtime`, `spoken_languages`, `cast`, and `director`. These span both numerical variables (e.g., `budget`, `runtime`) and categorical features (e.g., `genres`, `director`). Features such as `vote_average`, `vote_count`, and `popularity` were deliberately excluded, as they are only available after a film's release and would constitute data leakage. The target variable is the `revenue` column, representing the total global box office receipts.

3 Proposed Solutions & Algorithms

In this section, we describe the standard approaches for movie revenue prediction and select three models to implement and compare.

Linear Regression: This will serve as our baseline model. It is a simple and interpretable approach that assumes a linear relationship between input features (e.g., `budget`) and the target revenue. It allows us to evaluate the necessity of more complex architectures.

Random Forest: This ensemble learning method is widely used for its ability to handle non-linear dependencies and its robustness against outliers. By combining multiple decision trees, it provides a more accurate estimation than simple linear models.

Multilayer Perceptron (MLP): A Multilayer Perceptron is a fully connected feedforward neural network that learns non-linear relationships through successive layers of neurons and activation functions, trained via backpropagation. We include it to experimentally evaluate whether a neural network can compete with tree-based methods on tabular data, despite typically requiring more preprocessing and hyperparameter tuning.

4 Preprocessing

The Movies dataset is composed of 7 **csv** files, to build the revenue prediction model only the following three will be needed:

- **credits.csv** – Contains the crew and cast of each movie.
- **keywords.csv** – Contains keywords related to each movie.
- **movies_metadata.csv** – Contains a plethora of information about each movie.

After analyzing each of these files there were some aspects that needed to be taken care of.

credits – This file has a lot of information, most of all is not needed for this model, so out of this **only the director and the first five names** of the cast were kept. This is because the cast is organized in order of importance in the movies. In reality, only cinephiles will look at the rest of the crew other than the director, and for the cast, only the bigger names influence the masses to watch a movie.

keywords – Every movie has around 5 to 10 keywords, and the amount of total unique keywords is in the thousands, this is way too much and most of them will not be relevant. In this phase of the project we chose to keep the **200 most common keywords**.

movies_metadata – This file is by far the most important, since it contains all the other features mentioned before. It does, however, have a big problem that might severely compromise the results for this project – **The budget and revenue features are largely incomplete**, out of ≈ 45000 movies, only ≈ 7000 have a values for the revenue – This is specially problematic, since it is impossible to train a model without the target value, this means that the dataset will inevitably need to be drastically reduced. And to complicate this even further, out of the ≈ 45000 movies, only ≈ 5000 have a budget value – this isn't as problematic as not having a revenue but it still affects our models, as it's possible to see in [Figure 1](#), this relation is expected to be one of the most influential in the revenue prediction problem. To counter this there are two options: the first is to remove every movie without a budget, this reduces our total movie collection to **5369 movies**; the second alternative is to calculate the mean budget from the movies with budget and impute it on the ones without, leaving **7380 movies** in the dataset.

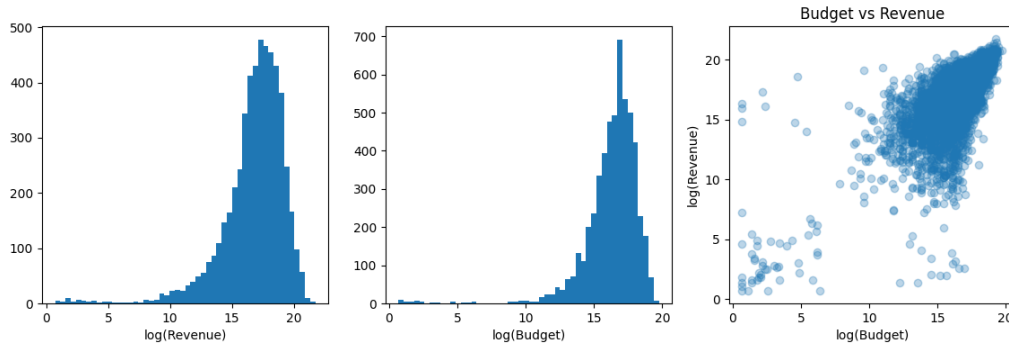


Figure 1: Relation between budget and revenue

It was decided to take both approaches and later keep the one with the best results.

5 Results

As previously mentioned, this project makes use of three different models – **Linear Regression**, **Random Forest** and **Multilayer Perceptron (MLP)**, all imported from the `sklearn` library. All models were trained on an 80/20 train/test split with `random_state=42` to ensure reproducibility. The target variable `revenue` and the feature `budget` were transformed using a logarithmic transformation (`log1p`) prior to training, as both distributions were heavily right-skewed. Predictions were converted back to USD using the inverse transformation (`expm1`) for evaluation. The models were evaluated using Root Mean Squared Error (RMSE) and the coefficient of determination (R^2).

5.1 Linear Regression

Linear Regression was implemented as a baseline model. It achieved an R^2 of **0.236** and an RMSE of **\$133,456,208**, meaning the model explains only 23.6% of the variance in movie revenue. This was expected, as the relationship between the selected features and revenue is inherently non-linear.

5.2 Random Forest

The Random Forest model was trained with `n_estimators=200`, `max_depth=None`, and `min_samples_split=5`, obtained via a 4-fold `GridSearchCV`. It achieved an R^2 of **0.460** and an RMSE of **\$112,154,509**, a significant improvement over the baseline. However, a large gap was observed between the cross-validation R^2 (0.889) and the test R^2 (0.460), suggesting the model is overfitting. Addressing this through stronger regularisation (e.g., limiting `max_depth` or increasing `min_samples_leaf`) will be a priority in the final stage.

5.3 Multilayer Perceptron

The MLP was trained with a (128, 64) architecture, ReLU activations, and a learning rate of 0.0001, with early stopping enabled. Both input features and the target variable were normalised using `StandardScaler` prior to training. The model achieved an R^2 of **-6.297** and an RMSE of **\$412,433,084**, performing significantly worse than both other models. This result is consistent with findings in the literature suggesting that tree-based methods tend to outperform neural networks on tabular data, particularly with limited dataset sizes.

5.4 Comparison

Table 1 summarises the performance of all three models on the test set.

Model	RMSE (USD)	R^2
Linear Regression	\$133,456,208	0.236
Random Forest	\$112,154,509	0.460
MLP	\$412,433,084	-6.297

Table 1: Model performance on the test set.

Random Forest is currently the best performing model. The next steps will focus on reducing overfitting in the Random Forest, further tuning the MLP architecture, and comparing results between the two dataset versions (with and without budget imputation).

6 Future Steps

For the final solution of this project, several improvements will be pursued to address the current under-performance of the models.

The first step will be to search for a more complete dataset. With only 5,000 to 7,000 samples after preprocessing, the dataset is insufficient given the number of features present. A larger dataset would allow the models, particularly the MLP, to learn more robust patterns.

The second step will be to further tune the hyperparameters of the existing models. For the Random Forest, the focus will be on reducing the observed overfitting by constraining `max_depth` and increasing `min_samples_leaf`. For the MLP, different architectures and learning rates will be explored. Additionally, both models will be evaluated on the two dataset versions – with and without budget imputation – to determine which approach yields better results.

The third step will be to experiment with at least one additional model, to evaluate whether alternative approaches can achieve better performance with the available data.

Finally, further feature engineering will be explored, such as reducing the number of keywords and experimenting with different target encoding strategies for `director` and `cast`. Maybe even reducing the number of features.

7 Bibliography

References

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